



Residential Society Resident and Maintenance Billing System | Presentation-I

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Project Overview

The system stores information about societies, blocks, flats, residents, maintenance bills, and payments in a structured relational database.

The main objective is to replace manual registers or disconnected spreadsheets with a centralized database that can:

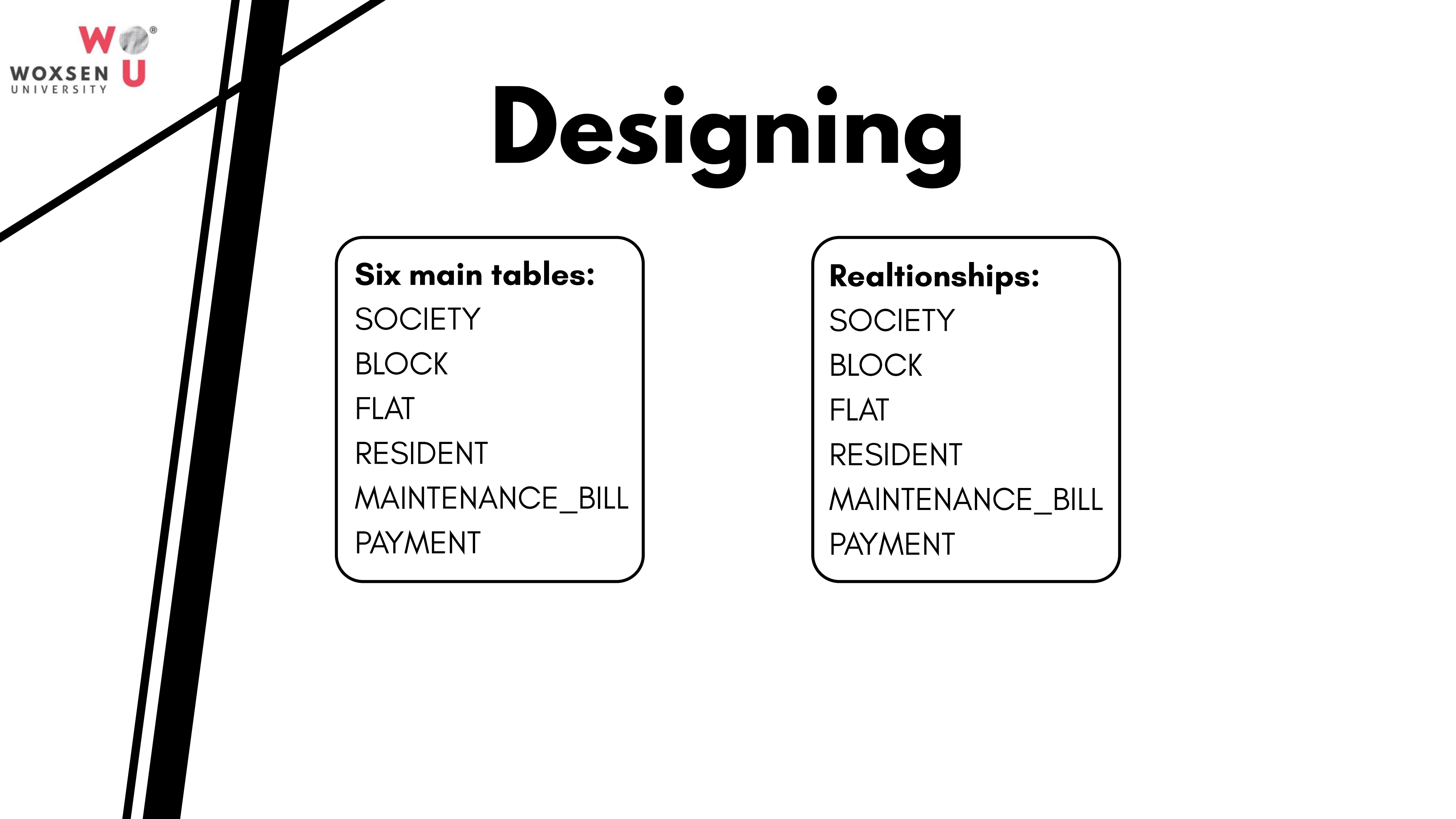
- Maintain details of society blocks and flats.
- Store resident information and their association with flats.
- Generate monthly maintenance bills.
- Record maintenance, water, parking, and late-fee charges.



Requirements

It must do the followign to classify as usable:

- Store society and block details.
- Manage flats and residents.
- Generate monthly maintenance bills.
- Record water, parking, and late charges.
- Record payments and payment methods.
- Track Paid / Partial / Unpaid bills.
- Maintain data security and integrity.



Designing

Six main tables:

SOCIETY
BLOCK
FLAT
RESIDENT
MAINTENANCE_BILL
PAYMENT

Realtionships:

SOCIETY
BLOCK
FLAT
RESIDENT
MAINTENANCE_BILL
PAYMENT

Implementation

Queries like these can be used to make to find various info:

- Who lives in a certain block and where
- Who has paid the maintainance bill and not
- How many flats we have

```
SELECT
    s.society_name,
    b.block_name,
    f.flat_number,
    r.resident_name,
    m.total_amount,
    p.amount_paid

FROM Society s

JOIN Block b
    ON s.society_id = b.society_id

JOIN Flat f
    ON b.block_id = f.block_id

JOIN Resident r
    ON f.flat_id = r.flat_id

JOIN Maintenance_Bill m
    ON f.flat_id = m.flat_id

JOIN Payment p
    ON m.bill_id = p.bill_id

WHERE r.resident_name = 'Rahul';
```

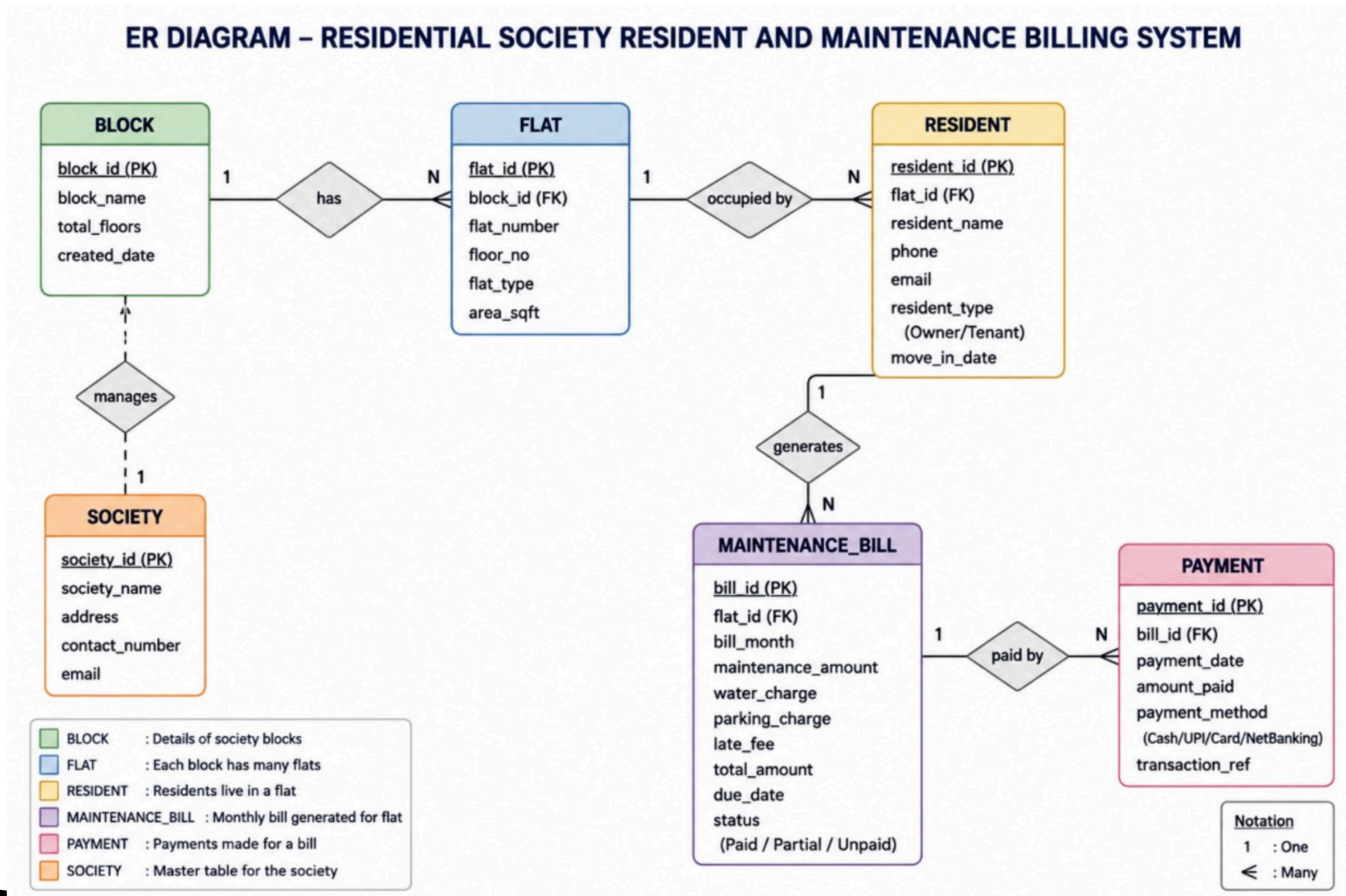
Business Rules

It must follow the following business rules:

- One society can have many blocks.
- One block can have many flats.
- One flat can have many residents.
- One flat can have many bills.
- One bill can have multiple payments.
- Every foreign key must reference an existing record.
- Bill status = Paid, Partial, or Unpaid.
- Total Amount = Maintenance + Water + Parking + Late Fee.
- Payment amount should not exceed the bill amount.



ER Diagram



society_id	society_name
1	Green Valley

block_id	society_id	block_name
10	1	Block A
11	1	Block B

payment_id	bill_id	amount_paid
5001	1001	3,800
5002	1002	2,000

flat_id	block_id	block_number
101	10	A-101
102	10	A-102
201	11	B-201

resident_id	flat_id	resident_name
501	101	Rahul
502	102	Amit

bill_id	flat_id	bill_amount
1001	101	3,800
1002	102	4,000

Thank You

